

Siddharth Singh

sms10221@nyu.edu ♦ +1 (929)689-4615 ♦ New York
[linkedin.com/in/isiddharthsingh](https://www.linkedin.com/in/isiddharthsingh) ♦ isiddharthsingh.netlify.app/ ♦ github.com/isiddharthsingh

EDUCATION

New York University, New York, NY

Master of Science, Computer Science

September 2023 - May 2025

CGPA: 3.72/4.0

Relevant Coursework: Data Science, Cloud Computing, Algorithms, Database Systems, Machine Learning, Information Security and Privacy, Big Data, Operating Systems, Open Source and Software Development, R in Finance

SRM Institute of Science and Technology, Chennai, India

Bachelor of Technology, Computer Science

July 2017 - May 2021

CGPA: 8.10/10

PROFESSIONAL EXPERIENCE

Arvo AI, New York, USA | Software Developer

December 2025 - Present

- **Built an AI-powered incident RCA platform** that autonomously investigates alerts, as measured by 240-step reasoning across 40+ tools and 4 LLM providers, by designing a **LangGraph** ReAct agent with context trimming and live streaming.
- Engineered a multi-cloud infrastructure system spanning **5 cloud providers** and **15+ monitoring platforms**, as measured by credential-isolated execution with STS role assumption and OAuth2, by designing a pluggable connector architecture.
- **Designed an alert correlation engine** that auto-groups related alerts into incidents, as measured by combining 3 weighted strategies (topology, time-window, similarity), by building a Memgraph-backed scoring system with 11-method discovery.
- Built a 3-collection RAG system with episodic **agent memory**, as measured by enabling hybrid search and automated **learning from past investigations**, by implementing Weaviate with heading-aware chunking and session-scoped indexing.
- Built an internal lead management platform automating end-to-end outreach, as measured by reducing messaging from 5-6 hours to 30-40 minutes (**~90% reduction**), by integrating HeyReach's API replacing a manual process.
- Developed a multi-agent orchestration agent that is designed to perform three separate hypotheses. Then, it uses the correlation engine to continue finding the root cause analysis in parallel before jumping to any conclusion.

Kamen Yotov, New York, USA | Software Developer

June 2025 - December 2025

- Developed an **AI-powered productivity assistant** integrating Slack, Trello, Gmail, and Google Calendar APIs, automating tasks and calendar sync which led to cut down manual coordination by 50% using Slack Bolt and Google API client.
- Designed a system with **automation** tools enabling task creation, board management, and checklists on Trello, while summarizing unread Gmail messages to Slack, boosting task tracking and visibility by **40%**.

New York Wealth Planning Group Inc., New York, USA | Software Developer

July 2025 - December 2025

- I was responsible for designing, developing, and maintaining the ISAC website. This work includes creating landing pages, webinar pages with registration and filtering features, WhatsApp group automation, and volunteer databases.
- The role requires building user interfaces, managing data flows, and applying coding skills to develop functional and responsive web applications.

Futeur AI, New York, USA | Software Developer

February 2025 - May 2025

- **Architected microservices application** with Next.js and CI/CD pipelines, delivering **40% performance gains** and **75% fewer deployment failures** while implementing Clerk OAuth for **65% faster sign-ups**.
- Led security innovation for FuteurSecure by implementing post-quantum cryptography (**CRYSTALS-Kyber/Dilithium**) with plug-and-play database integration, achieving **100% NIST compliance** and **85% vulnerability reduction**.
- **Developed enterprise-grade security monitoring platform** that integrates **OSSEC HIDS**, **Wazuh** and **Suricata IDS/IPS**, reducing false positives by 60% and increasing the accuracy of **threat detection by 45%**.
- **Engineered distributed logging system** using Hyperledger Fabric blockchain infrastructure with analytics dashboards, increasing customer retention by more than 70% through data-driven user engagement insights.

New York University, New York, USA | Web Developer

September 2024 - May 2025

- Develop and maintain a modern department website for New York University using a **Java-based backend** and a React frontend, optimizing functionality, load times, and responsive design to **increase site engagement by 20%**.
- Build personalized faculty websites with secure login features, custom layouts, and seamless integration to improve faculty **accessibility and visibility by 30%** and enhance the user experience for students and faculty.

Cognizant Technology Solutions, Hyderabad, India | Cloud Engineer

July 2021 - August 2023

- **Increased team efficiency by 35%** by designing and implementing **scalable GCP-based infrastructure** using **Compute Engine**, **Kubernetes Engine**, and automating workflows to streamline cloud operations.
- **Boosted deployment speed and reliability by 40%** through building robust **CI/CD pipelines** with **Cloud Build**, **Cloud Run**, and **Terraform**, ensuring seamless continuous integration and delivery for cloud-native applications.
- **Optimized cloud infrastructure** and data workflows by developing **ETL pipelines** with **Dataflow** and **BigQuery**, automating provisioning with **Kubernetes** and **Deployment Manager**, and enhancing scalability and uptime.
- Enhanced cloud security and compliance by architecting **IAM roles**, **VPC peering**, and **firewall configurations** in GCP, achieving **100% audit readiness** and reducing security vulnerabilities by 30%.

PROJECTS

CryptoStream AI | *Apache Kafka, Apache Spark, Cassandra, Streamlit, Grafana, Docker*

- Engineered a **real-time data pipeline** using Apache Kafka, Spark, and Cassandra, to process, transform, and store high-velocity streaming data from the Coinbase API, enabling **real-time analytics** and low-latency querying.
- Developed **interactive dashboards** with Streamlit and Grafana, integrating **machine learning models** like LSTM, ARIMA, VAR, and Moving Averages for forecasting, and deployed a scalable solution using Docker. [Github](#)

Taleweaver | *AWS Rekognition, OpenAI API, AWS EC2, AWS Lambda, DynamoDB, Cognito*

- Built an AI-driven application leveraging AWS Rekognition and OpenAI API for image recognition, natural language processing, and story generation, hosted on AWS EC2 with a scalable serverless backend using Lambdas.
- Streamlined user authentication and data management by implementing AWS Cognito, DynamoDB, and API Gateway, ensuring secure, low-latency operations and seamless RESTful API interactions. [Github](#)

Serverless Dining Concierge Chatbot | *AWS Lex, Lambda, API Gateway, DynamoDB, Elasticsearch*

- Designed and deployed a serverless, microservices-based Dining Concierge chatbot using AWS Lex, Lambda, API Gateway, and S3, leveraging Natural Language Processing to deliver personalized restaurant recommendations.
- Automated restaurant suggestion workflows by 50% through integration with SQS, Elasticsearch, DynamoDB, and SES, orchestrated with CloudWatch events for efficient email notifications and data retrieval. [Github](#)

SKILLS AND COMPETENCIES

Programming: Python, C, C++, Java, SQL, HTML, JavaScript, .Net, C#

Databases: Redshift, CloudSQL, Cloud Spanner, AlloyDB, BigQuery, BigTable, DynamoDB, RDS, CockroachDB, MongoDB, Postgres

Other Technologies: GCP, AWS, Docker, Kubernetes, Git, NumPy, Pandas, PyTorch, Redis, React, CircleCI, HashiCorp Vault, LiveKit

Certifications: Professional Cloud DevOps Engineer, Associate Cloud Engineer, Deep Learning